
Control Is Motion

A New Framework for Nonlinear Systems
That Move Through the World

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Chapter 1

Throwing Away the Target

A river does not flow toward the ocean because it knows where the ocean is. It flows because flowing is what rivers do.

— Adapted from Lao Tzu

1.1 The Wrong Question

Open any classical control textbook and you will find, within the first few pages, a picture like this: a system, an error signal, a reference point, and a controller whose entire purpose is to drive that error to zero. The implicit promise is seductive—*give me a destination and I will take you there*.

But consider the golf swing.

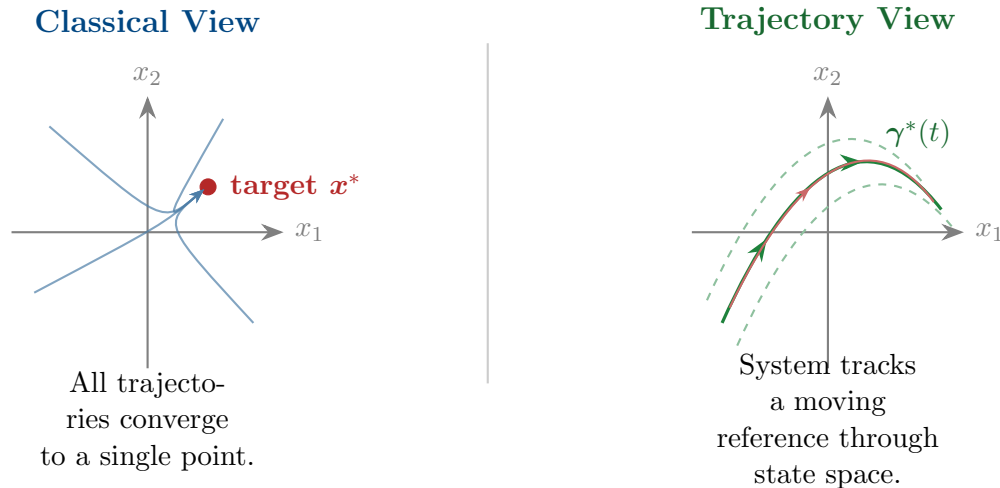
A skilled golfer does not steer the clubhead toward the ball the way a thermostat drives a room toward 72°F. There is no “setpoint” floating in space that the club is chasing. Instead, the golfer’s body executes a *motion*—a coordinated, whole-body trajectory through a high-dimensional configuration space. The clubhead arrives at the ball not because it was pointed at the ball, but because the entire motion was shaped so that, somewhere along its arc, the club happens to pass through the ball’s location at enormous speed.

Key Idea 1.1. The Central Thesis

Control is motion, not destination. The fundamental object of nonlinear control is not a setpoint, not an equilibrium, and not a fixed target. It is a *trajectory*—a curve through state space that the system is asked to follow, and around which disturbances are rejected. The quality of control is measured not by how close the system gets to a point, but by how faithfully it reproduces a desired motion.

This is not merely a philosophical reframing. It changes the mathematics. It changes what we optimize. It changes how we define stability. And it changes what “success” means in a controlled system.

The purpose of this book is to develop control theory from this trajectory-first perspective, to show that the classical setpoint framework is a special (and often misleading) case of a much richer theory, and to build the mathematical tools needed to design controllers for systems whose purpose is to *move*.



1.2 Why Setpoints Are a Special Case

Let us be precise about what we mean. Consider a nonlinear system

$$\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, \mathbf{u}), \quad \mathbf{x} \in \mathcal{X} \subseteq \mathbb{R}^n, \quad \mathbf{u} \in \mathcal{U} \subseteq \mathbb{R}^m, \quad (1.1)$$

where $\mathbf{f}$ is smooth and the state space $\mathcal{X}$ may carry additional geometric structure (it may be a manifold, as we will see in Chapter 3).

In the classical framework, the control objective is:

Find $\mathbf{u}(t)$ such that $\mathbf{x}(t) \rightarrow \mathbf{x}^$ as $t \rightarrow \infty$.*

The entire apparatus of Lyapunov stability, LQR, pole placement, and PID control is built around this question. But notice what this objective assumes:

- A1.** There exists a meaningful equilibrium $\mathbf{x}^*$ where $\mathbf{f}(\mathbf{x}^*, \mathbf{u}^*) = \mathbf{0}$.
- A2.** The system's purpose is to *reach and remain at* this equilibrium.
- A3.** Performance is measured by how quickly and smoothly the error $\mathbf{e}(t) = \mathbf{x}(t) - \mathbf{x}^*$ decays.

Now consider the systems this book is about:

- A golfer executing a swing—the body passes through a continuum of configurations and never “arrives” anywhere.
- A gymnast performing a floor routine—the motion *is* the performance.
- A biped walking—there is no single posture that constitutes “walking”; walking is a *limit cycle*, a periodic orbit through state space.
- A robotic arm tracing a weld seam—the task is defined by the path, not the endpoint.

For all of these systems, assumptions **A1**–**A3** fail. The correct control objective is fundamentally different:

Principle 1.1. The Trajectory Tracking Objective

Given a reference trajectory $\gamma^* : [0, T] \rightarrow \mathcal{X}$, find a feedback control law $\mathbf{u} = \mathbf{k}(\mathbf{x}, t)$ such that

$$\|\mathbf{x}(t) - \gamma^*(t)\| \leq \varepsilon(t) \quad \text{for all } t \in [0, T], \quad (1.2)$$

where $\varepsilon(t)$ is a prescribed tracking tolerance (the “tube width”), and the system is robust to disturbances $\mathbf{d}(t)$ satisfying $\|\mathbf{d}(t)\| \leq \delta$.

A setpoint is simply the degenerate case where $\gamma^*(t) = \mathbf{x}^*$ for all t and $T \rightarrow \infty$. The trajectory-tracking framework *contains* the setpoint framework, but not vice versa.

Definition 1.1. Trajectory Tube

Given a reference trajectory $\gamma^* : [0, T] \rightarrow \mathcal{X}$ and a tube radius function $\varepsilon : [0, T] \rightarrow \mathbb{R}_{>0}$, the **trajectory tube** is the time-varying set

$$\mathcal{T}(t) = \{ \mathbf{x} \in \mathcal{X} : \|\mathbf{x} - \gamma^*(t)\| \leq \varepsilon(t) \}. \quad (1.3)$$

A controller achieves **tube-invariance** if every solution starting in $\mathcal{T}(0)$ remains in $\mathcal{T}(t)$ for all $t \in [0, T]$.

This definition is the trajectory-centric replacement for Lyapunov stability. Instead of asking “does the system converge to a point?” we ask “does the system stay inside a moving tube?” The tube can be tight where precision matters (e.g., at ball impact in the golf swing) and loose where it does not (e.g., during the backswing).

1.3 Anatomy of a Motion: The Golf Swing as Prototype

We will use the golf swing throughout this book as a running example—not because golf is inherently more interesting than other motions, but because it compresses an extraordinary number of control-theoretic challenges into roughly 1.2 seconds:

- (i) **High dimensionality.** The human body has over 200 degrees of freedom. Even a simplified musculoskeletal model of the golf swing uses 20–40 generalized coordinates.
- (ii) **Extreme nonlinearity.** Joint torques couple through the full rigid-body dynamics, including Coriolis and centrifugal terms that dominate at high angular velocities.
- (iii) **Underactuation.** Once the club is released in the downswing, the wrist hinge “freewheels”—the golfer cannot directly control the club angle. The clubhead speed at impact comes from *passive dynamics*, not active torque.
- (iv) **Time-criticality with flexible timing.** The swing must produce the right state (clubhead position, velocity, and orientation) at impact, but the *exact* time of impact is not prescribed. The golfer is free to take the backswing slowly or quickly.

- (v) **Repeatability over optimality.** A touring professional does not execute the theoretically optimal swing. They execute a swing they can *repeat* 300 times in a tournament, under pressure, with minimal variance.

Example 1.1. From Setpoint Thinking to Trajectory Thinking

Suppose we model a simplified golf swing with generalized coordinates $\mathbf{q} = (\theta_{\text{torso}}, \theta_{\text{shoulder}}, \theta_{\text{arm}}, \theta_{\text{wrist}})^\top$ and their velocities $\dot{\mathbf{q}}$, giving state $\mathbf{x} = (\mathbf{q}, \dot{\mathbf{q}}) \in \mathbb{R}^8$.

Setpoint approach (wrong): Define a target state $\mathbf{x}^* = (\mathbf{q}_{\text{impact}}, \dot{\mathbf{q}}_{\text{impact}})$ and design a controller to drive $\mathbf{x}(t) \rightarrow \mathbf{x}^*$. This fails immediately: the system should not *stop* at $\mathbf{x}^*$. Impact is a waypoint, not a destination. The club must be accelerating *through* the ball, not decelerating toward it.

Trajectory approach (right): Define a reference motion $\gamma^*(s) : [0, 1] \rightarrow \mathbb{R}^8$ parameterized by a phase variable s (not clock time t). The controller's job is to ensure that the system traverses γ^* with the right sequencing and coordination, with a tight tube near the impact phase ($s \approx 0.7$) and a loose tube during setup ($s \approx 0$).

Point (v) deserves special emphasis. Classical optimal control asks: “What is the minimum-time or minimum-energy trajectory?” But this is the wrong question for biological and athletic systems. The right question is:

Principle 1.2. The Repeatability Principle

The best trajectory is not the one that maximizes performance on a single trial, but the one that maximizes *expected performance over many trials* in the presence of noise, fatigue, and uncertainty. Formally, we seek

$$\gamma^* = \arg \min_{\gamma} \mathbb{E} \left[\int_0^T \ell(\mathbf{x}(t), \mathbf{u}(t)) dt \right] + \lambda \text{Var}[J(\gamma, \mathbf{w})], \quad (1.4)$$

where J is the performance cost, $\mathbf{w}$ represents stochastic disturbances (motor noise, perceptual error), and $\lambda > 0$ penalizes variance.

This is why elite golfers do not all swing the same way. Each golfer finds a trajectory that is locally optimal *for their body and their noise characteristics*. Robustness is not an afterthought bolted onto an optimal controller—it is the primary design objective.

1.4 The Mathematical Landscape

To make the trajectory-first perspective rigorous, we need mathematical machinery that most introductory control courses skip entirely. This section provides a roadmap of the key concepts we will develop in subsequent chapters.

1.4.1 Phase Variables and Time Reparameterization

A crucial distinction separates *what* the system does from *when* it does it. Consider two executions of the same golf swing: one with a slow backswing and one with a fast backswing. The *spatial path* through configuration space may be identical, but the *time parameterization* differs.

We formalize this by introducing a **phase variable** $s(t) \in [0, 1]$ that monotonically increases from 0 (start of motion) to 1 (end of motion), with dynamics

$$\dot{s} = \alpha(s, \mathbf{x}), \quad (1.5)$$

where $\alpha > 0$ is the *phase velocity*. The reference trajectory is defined as a function of s , not t :

$$\boldsymbol{\gamma}^* = \boldsymbol{\gamma}^*(s), \quad s \in [0, 1]. \quad (1.6)$$

This decouples the *shape* of the motion from its *temporal execution*. The controller has two separate tasks:

- (a) **Path tracking:** keep $\mathbf{x}(t)$ close to the reference path $\boldsymbol{\gamma}^*(s(t))$, and
- (b) **Timing control:** regulate $\dot{s}$ to achieve the desired speed profile along the path.

This decomposition is impossible in the setpoint framework, where time and space are fused together.

1.4.2 Transverse Dynamics and Orbital Stability

When we shift from point stability to trajectory stability, the relevant notion of “closeness” changes. We do not care about the distance from $\mathbf{x}(t)$ to $\boldsymbol{\gamma}^*(t)$ at the same clock time—we care about the distance from $\mathbf{x}(t)$ to the *nearest point on the trajectory*.

Definition 1.2. Transverse Distance

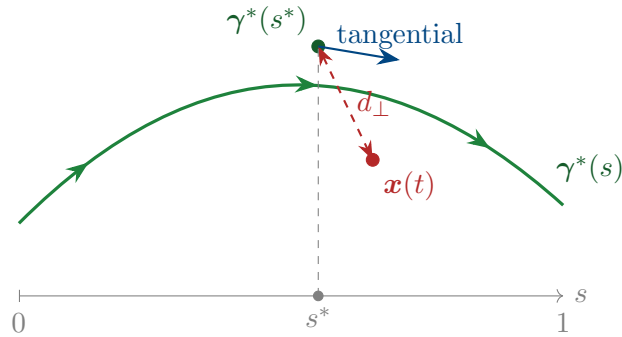
Let $\boldsymbol{\gamma}^* : [0, 1] \rightarrow \mathcal{X}$ be a smooth reference trajectory. The **transverse distance** from a state $\mathbf{x}$ to $\boldsymbol{\gamma}^*$ is

$$d_{\perp}(\mathbf{x}, \boldsymbol{\gamma}^*) = \min_{s \in [0, 1]} \|\mathbf{x} - \boldsymbol{\gamma}^*(s)\|. \quad (1.7)$$

The minimizing argument $s^*(\mathbf{x}) = \arg \min_s \|\mathbf{x} - \boldsymbol{\gamma}^*(s)\|$ is the **projection phase**, giving the “closest point on the reference.”

This leads to a decomposition of the state into two components:

- The **tangential component**: where the system is *along* the trajectory (measured by s).
- The **transverse component**: how far the system is *from* the trajectory (measured by $d_{\perp}$).



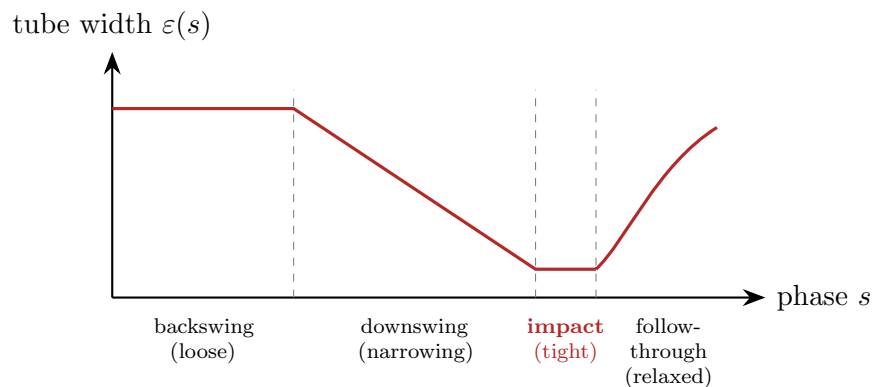
A system is **orbitally stable** around γ^* if the transverse distance $d_{\perp}(x(t), \gamma^*)$ remains small (or decays) even if the tangential phase $s(t)$ drifts relative to the nominal timing. This is exactly the right notion for the golf swing: we want the motion to follow the right *path* even if the golfer is slightly faster or slower than planned.

1.4.3 Funnel Control and Time-Varying Tubes

Combining the trajectory tube (Definition 1.1) with the transverse/tangential decomposition leads to what we call **funnel control**: the design of controllers that keep the system inside a prescribed time-varying tube around the reference.

The tube width $\varepsilon(s)$ is a design parameter. For the golf swing:

$$\varepsilon(s) = \begin{cases} \varepsilon_{\text{loose}} & s \in [0, 0.3] \quad (\text{address \& backswing}), \\ \varepsilon_{\text{loose}} - (\varepsilon_{\text{loose}} - \varepsilon_{\text{tight}}) \frac{s - 0.3}{0.4} & s \in [0.3, 0.7] \quad (\text{transition \& downswing}), \\ \varepsilon_{\text{tight}} & s \in [0.7, 0.8] \quad (\text{impact zone}), \\ \text{relaxed} & s \in [0.8, 1.0] \quad (\text{follow-through}). \end{cases} \quad (1.8)$$



This funnel profile encodes the physical insight that precision matters most at impact and least during setup. The controller “spends” its control authority where it matters, rather than trying to maintain uniform precision everywhere—a lesson that classical setpoint control cannot teach.

1.5 What Changes When Control Is Motion

Adopting the trajectory-first perspective changes nearly every aspect of control system design. Here we preview the major consequences, each of which will be developed in detail in subsequent chapters.

1.5.1 Stability Is Not Convergence—It Is Confinement

In the setpoint framework, stability means convergence: the state approaches the equilibrium and stays near it. In the trajectory framework, stability means *confinement*: the state remains inside the moving tube.

Definition 1.3. Trajectory Stability

A reference trajectory γ^* is **stable** under the closed-loop dynamics $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, \mathbf{k}(\mathbf{x}, t))$ if for every $\varepsilon > 0$ there exists $\delta > 0$ such that

$$d_{\perp}(\mathbf{x}(0), \gamma^*) < \delta \implies d_{\perp}(\mathbf{x}(t), \gamma^*) < \varepsilon \quad \forall t \in [0, T]. \quad (1.9)$$

It is **asymptotically trajectory-stable** if, additionally, $d_{\perp}(\mathbf{x}(t), \gamma^*) \rightarrow 0$ as the system evolves.

Note the subtle but critical difference: we use the transverse distance $d_{\perp}$, not the time-indexed error $\|\mathbf{x}(t) - \gamma^*(t)\|$. This allows the system to be ahead of or behind schedule without being considered “unstable.”

1.5.2 The Cost Function Is a Functional, Not a Function

In setpoint control, the cost is typically a function of the error $\mathbf{e}(t)$:

$$J_{\text{setpoint}} = \int_0^{\infty} (\mathbf{e}^{\top} \mathbf{Q} \mathbf{e} + \mathbf{u}^{\top} \mathbf{R} \mathbf{u}) dt. \quad (1.10)$$

In trajectory control, the cost is a **functional** that operates on the entire motion:

$$J_{\text{traj}}[\mathbf{x}(\cdot), \mathbf{u}(\cdot)] = \underbrace{\int_0^T w_{\perp}(s) d_{\perp}^2(\mathbf{x}(t), \gamma^*) dt}_{\text{tracking fidelity (transverse)}} + \underbrace{\int_0^T w_{\parallel}(s) (\dot{s} - \dot{s}_{\text{ref}})^2 dt}_{\text{timing fidelity (tangential)}} + \underbrace{\int_0^T \mathbf{u}^{\top} \mathbf{R}(s) \mathbf{u} dt}_{\text{effort}}, \quad (1.11)$$

where the weighting matrices $w_{\perp}(s)$, $w_{\parallel}(s)$, and $\mathbf{R}(s)$ are *phase-dependent*. This is what allows us to demand high precision at impact and low effort during the backswing.

1.5.3 Feedforward Is Not Optional—It Is Primary

In the setpoint framework, feedforward is a “nice to have” that reduces transient error. In the trajectory framework, feedforward is the *primary* control action. The reference trajectory

$\gamma^*(t)$ implicitly defines a feedforward control signal

$$\mathbf{u}_{\text{ff}}(t) = \mathbf{f}^{-1}(\dot{\gamma}^*(t), \gamma^*(t)), \quad (1.12)$$

where $\mathbf{f}^{-1}$ denotes the inverse dynamics. The feedback controller handles only the *deviations* from this feedforward signal:

$$\mathbf{u}(t) = \underbrace{\mathbf{u}_{\text{ff}}(t)}_{\text{executes the motion}} + \underbrace{\mathbf{u}_{\text{fb}}(\mathbf{x}(t), t)}_{\text{corrects deviations}}. \quad (1.13)$$

For the golf swing, the feedforward component accounts for approximately 90% of the total joint torques. The feedback controller provides the remaining 10% of correction. A control design that treats everything as feedback is asking the feedback loop to do ten times more work than necessary.

1.5.4 Underactuation Creates Geometry, Not Just Constraints

Many moving systems are underactuated: they have fewer control inputs than degrees of freedom. In the setpoint framework, underactuation is an obstacle—it means we cannot independently control all states. In the trajectory framework, underactuation creates *geometric structure* that we can exploit.

The set of states reachable from a given configuration through passive dynamics forms a submanifold of the state space. A well-designed trajectory *rides along* this manifold, using the passive dynamics to generate motion that active torques alone could not produce. This is how the golf swing works: the wrist hinge releases passively, and the centrifugal acceleration of the downswing *automatically* squares the clubface through the double pendulum effect.

Proposition 1.1. *For a mechanical system with n degrees of freedom and $m < n$ actuators, the set of dynamically feasible trajectories lies on a submanifold of dimension at most $2m$ in the $2n$ -dimensional state space. Effective trajectory design requires understanding the geometry of this submanifold.*

We will make this precise in Chapter 5 using the language of constraint manifolds and their null-space structures.

1.6 A Roadmap for the Book

The remainder of this book develops the trajectory-first theory in a sequence of increasingly sophisticated layers:

Ch. 2 Curves in State Space. The geometry of trajectories: parameterization, arc length, curvature, and torsion in high-dimensional state spaces. Frenet–Serret frames for nonlinear systems.

Ch. 3 Configuration Manifolds. Why state space is often not $\mathbb{R}^n$: Lie groups, joint spaces, and the geometry of articulated bodies. The golf swing lives on $SO(3)^k$, not $\mathbb{R}^k$.

- Ch. 4 Orbital Stability and Transverse Linearization.** Rigorous definitions of trajectory stability. Linearization transverse to the orbit. Floquet theory for periodic motions. Moving Poincaré sections.
- Ch. 5 Underactuation and Passive Dynamics.** Constraint Jacobians, null-space control, and the exploitation of passive mechanical coupling. The double pendulum and the physics of the wrist release.
- Ch. 6 Trajectory Optimization.** Direct collocation, shooting methods, and pseudospectral methods for finding optimal reference trajectories. The interplay between optimality and repeatability.
- Ch. 7 Funnel Synthesis.** Designing time-varying tubes with guaranteed invariance. Sums-of-squares programming for funnel computation. Robust funnels under model uncertainty.
- Ch. 8 Phase-Variable Control.** Decoupling path shape from timing. Virtual constraints and hybrid zero dynamics for walking and running. Central pattern generators as phase oscillators.
- Ch. 9 Stochastic Trajectories and Motor Variability.** Signal-dependent noise in biological actuators. The minimum-variance theory of human movement. Why optimal trajectories are smooth.
- Ch. 10 Learning to Move.** Iterative learning control, policy search, and trajectory libraries. How to improve a motion over repeated trials without losing stability.
- Ch. 11 Case Study: The Complete Golf Swing.** Full application of the entire theory to a 15-DOF musculoskeletal model of the golf swing, from trajectory optimization through funnel synthesis to stochastic robustness analysis.

1.7 A New Language for Control

Before proceeding, let us establish the conceptual vocabulary that will replace the classical language throughout this book:

Classical term	→	Trajectory term
Setpoint / reference	→	Reference trajectory $\gamma^*(s)$
Error $e(t) = \mathbf{x} - \mathbf{x}^*$	→	Transverse deviation $d_\perp(\mathbf{x}, \gamma^*)$
Lyapunov stability	→	Orbital / trajectory stability
Convergence to equilibrium	→	Confinement to tube
Regulation	→	Path tracking + timing control
Step response	→	Trajectory tracking response
Steady-state error	→	Persistent transverse offset
Gain margin	→	Funnel robustness margin
Pole placement	→	Transverse eigenvalue assignment

1.8 Exercises

1.1. (Conceptual.) Consider a simple pendulum $\ddot{\theta} + \sin \theta = u$.

- (a) Identify the equilibria and classify their stability.
- (b) Now consider the task of making the pendulum execute a full revolution (pumping up from rest to continuous spinning). Explain why this task *cannot* be formulated as setpoint regulation.
- (c) Sketch the reference trajectory in the $(\theta, \dot{\theta})$ phase plane for one complete revolution. What is the natural phase variable?

1.2. (Mathematical.) For the system $\dot{x}_1 = x_2$, $\dot{x}_2 = -x_1 + u$, consider the reference trajectory $\gamma^*(t) = (\cos t, -\sin t)^\top$ (a unit circle in state space).

- (a) Verify that $u_{\text{ff}}(t) = 0$ produces exactly this trajectory.
- (b) Compute the transverse linearization about γ^* .
- (c) Design a feedback law u_{fb} that achieves asymptotic orbital stability.

1.3. (Design.) A simplified 2-DOF golf swing model has coordinates $\mathbf{q} = (\theta_{\text{arm}}, \theta_{\text{club}})$ with dynamics

$$\mathbf{M}(\mathbf{q})\ddot{\mathbf{q}} + \mathbf{C}(\mathbf{q}, \dot{\mathbf{q}})\dot{\mathbf{q}} + \mathbf{g}(\mathbf{q}) = \mathbf{B} \mathbf{u}, \quad \mathbf{B} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

This system is underactuated: only θ_{arm} is directly actuated.

- (a) Explain qualitatively how the club can accelerate through passive dynamics even though no torque is applied at the wrist.

- (b) Define a phase variable s and sketch a funnel profile $\varepsilon(s)$ appropriate for this system.
- (c) Discuss why a setpoint controller applied at impact ($s = 0.7$) would produce the wrong motion.

1.4. (Philosophical.) A thermostat, an autopilot, and a dancer's body are all "controlled systems." Classify each according to whether its primary control objective is (i) setpoint regulation, (ii) trajectory tracking, or (iii) something else entirely. Justify your classification.

The river does not arrive at the ocean and stop.

It becomes the ocean.

The swing does not arrive at impact and stop.

It becomes the follow-through.

Control is motion.